

Smart Home Automation System Using Internet of Things

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Abstract---Due to the rapid development in the field of automation and IoT, human life is becoming more convenient and efficient. In the present scenario, automated systems are increasingly preferred over manual systems. With the growth of internet connectivity and smart devices, the Internet of Things (IoT) has become an essential technology for modern automation applications. In this paper, the design and implementation of a Smart Home Automation system using ESP32-CAM are presented. The system enables users to control home appliances such as lights through a Wi-Fi-hosted web page. An LED and pushbutton are used as the controlled load and manual switch respectively. The web interface displays the current status of each switch in real time. Additionally, an LDR (Light Dependent Resistor) sensor is integrated for automatic operation. When switched to auto mode, the system monitors ambient light intensity and controls the LED accordingly. This approach provides efficient energy management, convenience, and flexibility to the user. The proposed system can be accessed by multiple devices such as smartphones and laptops within the same network. It can also be expanded to include additional sensors and security modules for future applications.

Keywords—Smart Home Automation (SHA), Internet of Things (IoT), ESP32-CAM, LED Control, LDR Sensor, Auto/Manual Mode

I. INTRODUCTION

A "Smart Home Automation" system is a system that allows consumers to control various appliance of the

different kind used and also makes controlling of home appliances a very easier and saves energy. Nowadays automation system for home and building are used quite often [1]. On the other hand, they increase the comfort particularly when everyone is busy with their work. "Home automation systems" installed in houses apart from allowing centralized controlling of ventilation, heating, air condition and lighting [2], increase comfort as well.

So, they provide an overall reduction in cost and also useful in saving the energy which is certainly the main problem today. Wired communication forms the backbone of many well established smart home system. But costs are very high for already existing home automation systems. In contrast home automation systems can be of great use for home environment. Also, in the past Wi-Fi system have been used all around [4].

In home networking, Wi-Fi have become more common in current years. There are many advantages of smart homes and building automation systems that do not only use a wired network [5].

The advantages of Smart Home Automation usually fall into the categories, including energy savings, home safety, user convenience, and better control.

The following main attributes of Smart Home Automation are as follows [6]:

- Energy Savings: Smart light bulbs, fans, and switchboard uses less energy as well as reduces utility cost.
- Home Safety: Many home automation system provides the best technologies for home security. In order to make home safer and secure users buy these devices. Automated system for lighting and sensors detecting motion aid people in entering in a room and walking at night.
- User Convenient: Home automation provides great convenience to end users as they perform role task autonomously. For instance, you could use sensors indoors for switching on/off lights as required.
- Better Control: This system provides better control to end users within their home. One can easily know the happenings inside their home with this system.
- Comfortable Atmosphere: All Connected devices around our home can aid to develop a convenient and comfortable environment by providing reliable lighting temperature and sound.
- Provide Peace of Mind: This technology may provide consumers peace of mind. The prototype of SHA is depicted in Figure 1.



Fig. 1. Prototype of Smart Home Automation System

II. INTERNET OF THINGS (IoT)

The IOT can connect numerous kinds of things, for example, personal computer, tablets, and smart phones to the internet and also facilitates a new kind of communications between two or more things, or between things and people. Any object with an "IP address" can transmit data over a network successfully because of IoT. Many problems get their solution through internet, which can connect devices even in remote areas. The IoT improves the living standard of life with its inovating ideas. The development of IoT will revolutionise sectors like

automation, transportation, health centre, energy and houses [7][8].

In this paper, Wi-Fi technology is used as a Internet of Things. Wi-Fi is a famous wireless networking paradigm and uses radio waves for providing high speed internet. A common mistake is made when Wi-Fi is considered as short for wireless fidelity, but, infact, Wi-Fi simply means IEEE 802.11x. Wi-Fi, as defined by the Wi-Fi alliance, which is an organisation owning the Wi-Fi registered trade mark, is any wireless LAN (WLAN) product based on IEEE 802.11 standards. The different version of Wi-Fi are given by various IEEE 802.11 protocol standards, with the different radio technologies affecting their ranges, radio bands, and speeds achieved..

III. SYSTEM ANALYSIS

The Smart Home Automation (SHA) system consists of a hardware implementation that has the capability to control different home appliances within the user's home using ESP32-CAM Wi-Fi technology. Along with the controlling of appliances, different components such as an LED, pushbutton, and LDR sensor are used for automation and monitoring purposes. The LDR sensor detects the light intensity around it and accordingly controls the LED when the mode is switched to automatic, while in manual mode the light can be controlled through the pushbutton or the ESP32-hosted web interface is shown in Figure 2.

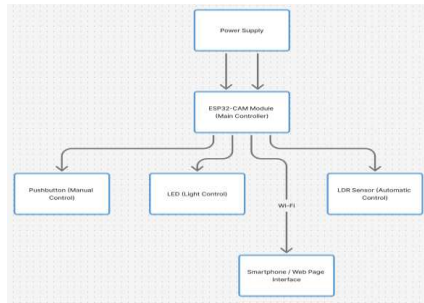


Fig. 2. Block Diagram for Smart Home Automation System

TABLE.I. LIST OF COMPONENTS

S.No	Components	Voltage (Volt)	Current (Amp)	Quantity (No.)
1	Power Supply	12V	1A, 2A	2
2	Voltage Regulator	3.3V	-	1
3	LED	5V		1
4	ESP32-CAM Module	3.3V	-	1
5	Pushbutton Switch	5V		1
6	LDR Sensor	5V	-	1
7.	Breadboard and Jumper wires	-	-	1

IV. HARDWARE DESCRIPTION

A. Power Supply

A power supply is a device responsible for providing electrical power to the entire circuit. Its primary function is to convert the input current from the source into the desired voltage and frequency required by the components. In the proposed system, a 12V, 1A adapter is used to power the ESP32-CAM module, while a 5V regulated supply is derived using a voltage regulator for other low-power components such as the LED, pushbutton, and LDR sensor. Figure 3 shows the circuit for the 5V regulated power supply.

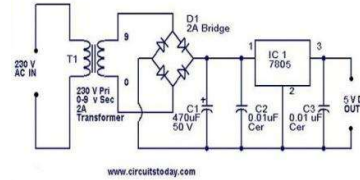


Fig. 3. Power Supply of 5V

B. ESP32-CAM Module

The ESP32-CAM is a compact microcontroller module with built-in Wi-Fi and camera support. In this system, it acts as the main control unit and web server for hosting a user interface that allows switching operations through a webpage. It receives input from the pushbutton and LDR sensor and controls the LED accordingly. The ESP32-CAM operates at 3.3V and communicates over Wi-Fi, enabling real-time monitoring of switch status. The different pin diagram of ESP32-CAM is shown in Fig 4



Fig. 4. Pin Diagram of ESP32-CAM

TABLE.II. PIN DESCRIPTION OF ESP 32- CAM

PIN NO	PIN NAMES	Usage
1	5V	Connected to 5V power input
2	GND	Connected to ground of the circuit
3	IO12	General-purpose I/O pin
4	IO13	General-purpose I/O pin
5	IO15	General-purpose I/O pin
6	IO14	General-purpose I/O pin
7	IO2	General-purpose I/O pin
8	IO4	Used for connecting LDR
9	3.3V	Connected to regulated 3.3V supply
10	IO16	General-purpose I/O pin
11	IO0	Used for flashing the code
12	GND	Connected to ground of the circuit
13	VCC	Power Supply
14	IO3	Receives serial data
15	IO1'	Transmits serial data
16	GND	Connected to ground of the circuit

C. LED

An LED is used as the primary load in this system to represent the lighting control in a smart home. It operates on a 5V DC supply and can be turned ON or OFF either manually through the pushbutton or automatically based on the LDR readings. The LED state is also reflected on the ESP32-hosted webpage, providing real-time feedback to the user.

D. Pushbutton Switch

The pushbutton serves as a manual input device that allows users to control the LED directly. When pressed, it toggles the LED state, and the same status is updated on the web interface. It provides flexibility for manual operation when the system is not in automatic mode.

E. LDR Sensor

The LDR (Light Dependent Resistor) is used for detecting the ambient light intensity. In automatic mode, the ESP32-CAM reads the LDR sensor value and turns the LED ON when the surrounding light level is low and OFF when it is bright. This enables efficient energy saving and smart lighting control. Figure 5 shows the working principle of the LDR Sensor.

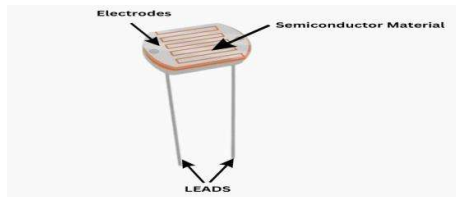


Fig. 5. LDR Sensor

F. Voltage Regulator

A voltage regulator which maintains a fixed voltage level automatically is used in this system. A voltage regulator may use a negative feedback or a simple feed-forward design. It may use electronic or an electromechanical mechanism. Based on the design, it can be used to operate one or more AC or DC voltage. The pin description for 7805 voltage regulator is shown in Figure 6.



Fig. 6. 7805 Voltage Regulator IC pin diagram

G. Breadboard and Jumper Wires

A breadboard and jumper wires are used for connecting all components in the circuit. The breadboard allows for easy testing and modification of the circuit without soldering, while jumper wires help establish the required electrical connections between the ESP32-CAM, LDR, LED, and pushbutton.

V. SOFTWARE DESCRIPTION

In the presented system, different software tools are used for programming and controlling the Smart Home Automation setup. The IDE (Integrated Development Environment) used is the Arduino IDE, which is an open-source platform for writing, compiling, and uploading programs to the ESP32-CAM module. The web interface for controlling the home appliances is created using HTML and hosted directly on the ESP32-CAM, allowing users to access and operate devices through a Wi-Fi connection. This interface also displays the current status of the switch in real time. The flow diagram for the Smart Home Automation process is shown in Fig 7.

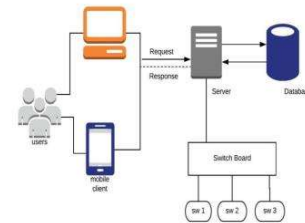


Fig. 7. Flow Diagram of Proposed Automation System

VI. IMPLEMENTATION SETUP

The implementation for the Smart Home Automation (SHA) system is carried out in the form of hardware. The setup consists of different hardware components connected through the ESP32-CAM module, which serves as the main controller and Wi-Fi interface for operating home appliances. The system is designed for both manual and automatic control of lighting using an LED, pushbutton, and LDR sensor. In Figure 8, the hardware connections of the setup are shown, where all components are interconnected through the ESP32-CAM module. The LED can be switched ON or OFF using the hosted web page or manually through the pushbutton. The complete hardware setup is divided into two cases. In the first case, manual mode operation using the pushbutton is shown. In the second case, automatic mode using the LDR sensor is demonstrated. Figure 9 shows the web page interface developed for controlling the light through Wi-Fi.

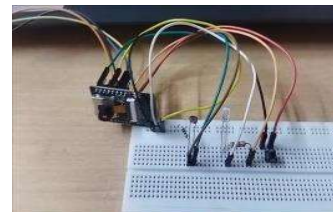


Fig.8. Hardware Implementation of Proposed Smart Home Automation System

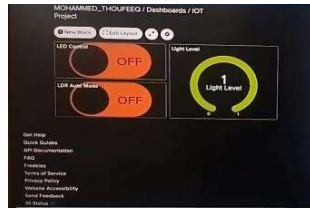


Fig.9 Software Implementation of Proposed Smart Home Automation System

VII. RESULTS AND DISCUSSION

The Smart Home Automation (SHA) system was implemented successfully using the ESP32-CAM module. The system operation is divided into two cases for better understanding, as described below.

CASE I: Manual Control of Home Appliances

In this mode, the LED can be controlled manually through the pushbutton or through the hosted web page. The webpage allows users to switch the light ON or OFF and view the real-time status of the switch from any device connected to the same Wi-Fi network. This provides convenience and efficient control of home appliances while reducing energy wastage.

CASE II: Automatic Control Using LDR Sensor

In this case, the LDR sensor monitors the surrounding light intensity and automatically turns the LED ON when the ambient light is low and OFF when it is bright. This feature enables energy-efficient smart lighting without the need for user intervention. The system responds quickly to changes in light levels, demonstrating effective automatic control.

The results confirm that the ESP32-CAM-based Smart Home Automation system performs reliably in both manual and automatic modes, providing flexible, low-cost, and user-friendly control of household lighting.

VIII. CONCLUSION

The Smart Home Automation (SHA) system has been tested successfully using the ESP32-CAM module with Wi-Fi connectivity, accessible through mobile phones, tablets, and laptops. The system not only controls home appliances such as lights but also provides monitoring features through sensors for automation. The integration of the LDR sensor enables automatic operation based on light intensity, while manual control is achieved through a pushbutton and the hosted web interface. Thus, this work demonstrates the prototype and implementation of a smart home automation system using Wi-Fi technology. The setup can be further expanded to control more appliances and include additional sensors for improved automation, monitoring, and energy management in future applications.

REFERENCES

- [1] Jasmine Guth, et. al., "A Detailed Analysis of IOT Platform Architectures: Concepts, Similarities, and Differences", Springer-Verlag, 2018
- [2] Majid Al Kuwari, Abdulrahman Ramadan, et. al., "Smart-Home Automation using IOT-based Sensing and Monitoring Platform", IEEE 12th International Conference on Compatibility, Power Electronics and Power Engineering, 2018
- [3] Kishore. P. T. Veeramankandasamy, K. Sabbath and S. Veerakumar, "Internet of Things based Low-Cost Real-Time Home Automation and Smart Security System", International Journal of Advanced Research in Computer and Communication Engineering, Volume 6, Issue 4, 2017
- [4] Smart Security System", International Journal of Advanced Research in Computer and Communication Engineering, Vol. 6, Issue 4, April 2017
- [5] SeungChul Son, NakWoo Kim, ByungTak Lee, Chae Ho Cho, et. al., "A Time Synchronization Technique for CoAP based Home Automation Systems", IEEE Transactions on Consumer Electronics, February 2016
- [6] Dhakad Kunal, DhakeTushar, UndegaonkarPooja, ZopeVaibhav, Vinay Lodha, "Smart Home Automation using IoT", International Journal of Advanced Research in Computer and Communication Engineering, Vol. 5, Issue 2, February 2016
- [7] Chi Shang Shih, Jyun He Choum, Niels Reamers, "Designing CPS/IOT applications for smart buildings and cities", IET Cyber-Physical Systems: Theory& Applications, 2016
- [8] Muhammad Asadullah, Ahsan Raza, "An Overview of Home Automation Systems", Transactions on Consumer Electronics IEEE, 2016
- [9] Nathan David, Anaphor Chime, Around Ugochukwu, Edoga Obinna, "Design of a Home Automation System Using Arduino", International Journal of Scientific & Engineering Research, Volume 6, 2015
- [10] Vinaysagar K. N, Kusuma S. M., "Home Automation Using Internet of Things", International Research Journal of Engineering and Technology, 2015
- [11] Remit Hilary, et. al., IOT Based Smart Home with Real-Time EMetering using controller", Annual IEEE India Conference, 2015
- [12] Somlyay Mahakam, R. Ramaswamy, Siddharth Tripathi, "Internet of Things (IOT): A Literature Review", Journal of Computer and Communications, 2015
- [13] Nathan David, Abafor Chime, et. al., "Design of a Home Automation System using Adriano", International Journal of Scientific & Engineering Research, 2015
- [14] Shopan Dey, Ayon Roy, "Home Automation Using Internet of Thing", International Research Journal of Engineering and Technology, 2015
- [15] Federico Viani, Fabrizio Robol, Alessandro Polo, "Wireless Architectures for Heterogeneous Sensing in Smart Home Applications: Concepts and Real Implementation", Proceedings of the IEEE, vol. 101, Issue: 11, Nov. 2013
- [16] Takeshi Yashiro, Shinsuke Kobayashi, Noboru Koshizuka, et. al., "An Internet of Things (IOT) Architecture for Embedded Appliances", IEEE Region 10 Humanitarian Technology Conference, 2013
- [17] Ahmed ElShafee, Kari Alana Hammed, "Design and Implementation of a Wife Based Home Automation System", World Academy of Science, Engineering and Technology, 2012
- [18] T. Kowatsch and W. Maass, "Critical Privacy Factors of Internet of Things Services: An Empirical Investigation with Domain Experts", Knowledge and Technologies in Innovative Information Systems, Lecture Notes in Business Information Processing, Vol. 129, Springer, Dordrecht, 2012
- [19] D. Kozlov, J. Veijalainen, and Y. Ali, "Security and Privacy Threats in IoT Architectures", Proc. of the 7th Int. Conf. on Body Area Networks, 2012.
- [20] Baoan Li, Jianjun Yu, "Research and application on the smart home based on component technologies and Internet of Things", Issued in Elsevier, 2011
- [21] R. van Kranenburg, E. Anzelmo, A. Bassi, D. Caprio, S. Dodson, and M. Ratto, "The Internet of Things", Proc. of the First Berlin Symposium on Internet and Society, 2011.